SPE-229380-MS

Artificial Intelligence Enhanced Prediction of Water Loading in Gas Wells: A Data-Driven Alternative to Turner's Model

Aashish Dheeraj Dadwani and Gaurav Hazarika, Pandit Deendayal Energy University

This paper was prepared for presentation at the ADIPEC held in Abu Dhabi, UAE, 3 – 6 November 2025.

Abstract

This study aims to develop an AI-enhanced, optimized empirical model for predicting critical gas rates to prevent water loading. By integrating real–field data and advanced machine learning techniques, a new Turner–style equation will be formulated. The model will dynamically adapt to varying well conditions, outperform traditional methods, and enable proactive, real–time management of deliquification strategies in complex environments. High- resolution well datasets, including bottomhole pressure, wellhead pressure, gas properties, and production rates, were extracted of a well X were taken and rigorously preprocessed. Variance analysis and permutation entropy were applied to detect early–stage flow regime transitions associated with liquid loading onset. Polynomial feature transformations captured nonlinear and cross–variable interactions among key operational parameters. A supervised learning framework with polynomial regression achieved 92% accuracy in predicting critical gas rates, representing a 28% improvement over conventional Turner–based methods. The final Turner–style AI–enhanced model dynamically adapts to changing multiphase flow regimes , enabling proactive, real–time optimization of deliquification strategies in Production systems.The polynomial regression approach , incorporating second–order and interaction terms between flowing bottomhole pressure (Pwf) and gas density (ρg),enabled the model to accurately capture complex nonlinear behaviors in multiphase offshore environments. Minimal deviation between predicted and actual critical gas rates was observed, confirming the model's robustness across varying well conditions. The flexible structure of the AI–based model allows seamless integration into digital twin platforms and real–time surveillance systems, enabling dynamic monitoring and proactive management of water loading risks. By adapting to fluctuating operational parameters, the model supports optimization of deliquification strategies, artificial lift planning, and well intervention scheduling. Overall, the AI–driven predictive framework presents a major advancement over conventional methods, offering a reliable, data– driven tool for enhancing gas production stability and extending well life in challenging production environments. Unlike traditional models that primarily focus on droplet or film reversal theories, this study introduces a hybrid AI– based approach that dynamically learns from multiphase behaviors. By integrating entropy-based instability detection and real–time data patterns, the developed model not only predicts the critical rate but also evolves adaptively to changing well conditions, ensuring sustained production optimization.

Introduction

The accurate prediction of critical gas rates is paramount for efficient production operations in gas wells, particularly in mitigating liquid loading, a pervasive issue that can severely impede gas flow and lead to premature well abandonment. Traditional methods, such as Turner's model, often rely on simplified assumptions that may not fully capture the complex multiphase flow dynamics observed in diverse field conditions, necessitating advanced predictive tools (Hosseini & Akilan, 2023). This study proposes an artificial intelligence-enhanced, data-driven empirical model to predict critical gas rates, offering a robust alternative to conventional approaches. This approach integrates real-field data and advanced machine learning techniques to formulate a new Turner-style equation that dynamically adapts to varying well conditions, thereby outperforming traditional methods (Pennel et al., 2018).

The model's adaptability ensures proactive, real-time management of deliquification strategies, even in complex operational environments (Masoudi et al., 2020). High-resolution well datasets, encompassing parameters like bottomhole pressure, wellhead pressure, gas properties, and production rates, were extracted from well X and subjected to rigorous preprocessing to ensure data quality and consistency. Variance analysis and permutation entropy were subsequently employed to detect early-stage flow regime transitions indicative of liquid loading onset, enhancing the model's predictive capabilities. Polynomial feature transformations were applied to capture nonlinear and cross- variable interactions among key operational parameters, leading to a supervised learning framework with polynomial regression that achieved 92% accuracy in predicting critical gas rates.

This performance represents a notable 28% improvement over conventional Turner-based methods, underscoring the benefits of an AI-driven approach in capturing complex reservoir behaviors (Liu & Pyrcz, 2023). The resulting AI- enhanced model, structured in a Turner-style format, dynamically adapts to changing multiphase flow regimes, facilitating proactive, real-time optimization of deliquification strategies within intricate production systems. The polynomial regression approach, specifically incorporating second-order and interaction terms between flowing bottomhole pressure and gas density (ρg), enabled the model to accurately capture complex nonlinear behaviors prevalent in multiphase offshore environments. Minimal deviation between predicted and actual critical gas rates was observed, confirming the model's robustness across varying well conditions.

Background

The flexible structure of this AI-based model allows seamless integration into digital twin platforms and real-time surveillance systems, enabling dynamic monitoring and optimization of production operations (Srikonda et al., 2020) (Aghito & Bjørkevoll, 2020). This integration streamlines logistics, minimizes maintenance costs, and refines decision-making processes by providing actionable insights derived from continuous data analysis (Li et al., 2024). The application of advanced artificial intelligence and machine learning methods, including deep learning, has shown promise in addressing the complex challenges of oil and gas production, offering superior analytical capabilities compared to traditional statistical methods (Arinze et al., 2024) & (Hosseini & Akilan, 2023).

These AI techniques excel at processing the vast quantities of high-resolution data generated in modern oil and gas operations, thereby enhancing predictive accuracy and operational efficiency (Chen et al., 2025). This paradigm shift towards AI-driven solutions is transforming the oil and gas industry by providing novel approaches for optimizing production dynamics, developing robust planning, and identifying residual oil, thereby revolutionizing the management of complex upstream processes (Hong et al., 2020). Furthermore, AI investments have demonstrated a significant return on investment within the oil and gas sector, yielding an average 32% ROI, alongside reductions in expenses and accelerated time-to-market for new products and services (Deif & Vivek, 2022).

The oil and gas industry is undergoing a significant transformation due to the integration of AI and machine learning, which enable more efficient, cost- effective, and safer operations by analyzing large datasets to derive actionable insights for data-driven decisions (Gowekar, 2024). This evolution is driven by the industry's need to tackle inherent uncertainties and capital-intensive processes, particularly in the upstream segment (Koroteev & Tekić, 2020). The ability of AI and ML to analyze large datasets and learn from historical trends is crucial for improving decision-making in the oil and gas industry, especially given the continuous generation of vast amounts of data (Sircar et al., 2021). This allows for enhanced performance prediction, optimized drilling operations, and improved reservoir management through the identification of subtle patterns that human analysis might overlook (Arinze et al., 2024) (Sircar et al., 2021).

The increasing complexity and volume of data within the upstream sector necessitate sophisticated analytical tools, making AI and machine learning indispensable for optimizing exploration, drilling, and production activities (Sircar et al., 2021). Such advancements lead to more precise forecasting of reservoir behavior, more efficient resource extraction, and reduced operational risks. The growing adoption of AI in the oil and gas industry is also driven by the need for sustainable practices and increased competitiveness, alongside the continuous challenges faced by the sector (Waqar et al., 2023).

This integration of AI is particularly vital given the industry's susceptibility to volatile market conditions, stringent environmental regulations, and the imperative for enhanced efficiency and safety (Wang et al., 2025). The application of artificial intelligence methods for identifying and predicting complications in the oil and gas sector is becoming increasingly popular and economically viable, as these methods can detect physically hidden processes and phenomena (Chernikov et al., 2020). This includes the use of AI to analyze factors like flow rates and pressures, which are critical for preventing issues such as liquid loading in gas wells (Sircar et al., 2021) (Gowekar, 2024). These AI-driven predictive capabilities not only enhance operational efficiency but also contribute significantly to environmental sustainability by optimizing resource utilization and minimizing the environmental footprint of extraction activities (Waqar et al., 2023).

Problem Statement

Despite these advancements, the industry still grapples with challenges such as unpredictable geological conditions, fluctuating market demand, and investment risks, which necessitate continuous innovation in predictive modeling (Chen et al., 2025). This underscores the critical need for more sophisticated, data-driven approaches to forecast and mitigate operational challenges like water loading, which can severely impede gas well productivity. Traditional empirical models, while foundational, often lack the adaptability and predictive power required to accurately model the complex, dynamic multiphase flow phenomena observed in diverse well conditions (Hong et al., 2020) (Kuang et al., 2021).

The proposed model will integrate advanced polynomial regression with real- field datasets, including bottomhole pressure, wellhead pressure, gas properties, and production rates, to accurately capture complex nonlinear interactions and enhance predictive robustness. This comprehensive data-driven strategy aims to improve the accuracy of crucial gas rate predictions, thereby alleviating the economic and operational challenges linked to liquid loading in gas wells. By applying variance analysis and permutation entropy, the model will adeptly detect early-stage flow regime transitions associated with liquid loading onset, ensuring timely intervention and optimization of production strategies. This enhanced model will represent a significant departure from static approaches, offering a flexible and adaptable solution for diverse operational scenarios. Furthermore, the methodology integrates polynomial feature transformations to effectively capture nonlinear and cross-variable interactions among key operational parameters, significantly improving the model's predictive capability.

Objectives

The primary objective of this study is to develop a robust, AI-enhanced empirical model for predicting critical gas rates, specifically designed to mitigate water loading in gas wells by leveraging real-field data and advanced machine learning techniques. This will involve formulating a new Turner-style equation that dynamically adapts to varying well conditions, thereby outperforming traditional methods and enabling proactive, real-time management of deliquification strategies in complex production environments. Specifically, this research aims to achieve several key objectives, including the accurate prediction of critical gas rates, the formulation of a dynamically adaptive model, and the significant improvement of existing methodologies for managing liquid loading in gas wells.

This includes a rigorous data preprocessing phase, employing variance analysis and permutation entropy to detect early-stage flow regime transitions indicative of liquid loading onset (Zhao et al., 2019). Furthermore, polynomial feature transformations will be applied to capture the intricate nonlinear and cross-variable interactions among critical operational parameters, enhancing the model's predictive accuracy. This rigorous approach will culminate in a supervised learning framework, employing polynomial regression to achieve high accuracy in critical gas rate predictions, representing a substantial improvement over conventional methods (Linden et al., 2015) (Cui, 2015) (Saputelli et al., 2005).

The final AI-enhanced model will dynamically adapt to changing multiphase flow regimes, enabling proactive, real-time optimization of deliquification strategies in production systems, while also assessing its seamless integration into digital twin platforms and real-time surveillance systems for dynamic monitoring (Lehr, 2020) (Hingerl et al., 2020) (Sarma et al., 2017) (Hossain & Ismail, 2013).

This integration will allow field engineers to access real-time predictions and scores, facilitating prompt action on model outputs (AlBar et al., 2019). The development of such a model is crucial for enhancing the longevity and efficiency of gas wells, aligning with the broader industry trend towards digitalization and predictive maintenance (Hajizadeh, 2019) (Liu & Jiang, 2022). This proactive approach, integrating artificial intelligence with real-time data from downhole sensors, significantly refines production optimization by providing accurate fractional composition and phase rates (Golenkin et al., 2020). This capability minimizes intervention costs and risks, reducing the well optimization cycle time and facilitating the continuous monitoring of well performance over extended periods, even through water breakthrough (Ajayi et al., 2019) (Golenkin et al., 2020).

This digital twin approach enables comprehensive monitoring of the entire life cycle of oil and gas assets, improving efficiency from design to operation control and optimization (Shen et al., 2021). This paradigm shift towards intelligent, data-driven solutions represents a significant advancement in the oil and gas industry, enabling more efficient and safer operations through enhanced predictive capabilities and real-time decision support (Gowekar, 2024) (Syed et al., 2020).

Contribution and Novelty

This study contributes significantly to the field by introducing a novel AI- enhanced Turner-style model that leverages advanced machine learning techniques to overcome the inherent limitations of traditional empirical models in predicting critical gas rates for water loading prevention. Specifically, this research innovates by integrating polynomial regression with high-resolution well datasets to accurately capture complex nonlinear behaviors in multiphase offshore environments, leading to a 28% improvement in prediction accuracy over conventional methods. This represents a significant advancement over existing methodologies, providing a more robust and adaptable solution for predicting critical gas rates under diverse operational conditions.

The flexible structure of this AI-based model allows for seamless integration into digital twin platforms and real-time surveillance systems, enabling dynamic monitoring and proactive management of deliquification strategies. Furthermore, the capability for real-time integration offers a pathway for enhanced operational efficiency and reduced intervention costs, positioning this model as a cornerstone for future advancements in reservoir management (D'Almeida et al., 2022) (Pandey et al., 2020). This approach offers significant value for maximizing asset value and minimizing risks by leveraging large datasets and advanced analytical techniques to forecast operational upsets (Cadei et al., 2018) (Marmier et al., 2017).

This predictive capability extends to identifying fault detections and preventing costly accidents in drilling and production (D'Almeida et al., 2022). It also provides a framework for transforming large volumes of raw data into actionable insights, facilitating critical thinking and informed decision-making in real-time operational environments (Bolen et al., 2018) (Temizel et al., 2017). This enables a proactive maintenance strategy by predicting equipment failures and optimizing intervention schedules, thereby preventing unexpected downtimes and maximizing production uptime. This augmentation of artificial intelligence with domain knowledge presents a robust solution for complex reservoir challenges, moving beyond mere data analysis to true knowledge discovery and systematic value creation (Castiñeira et al., 2018).

This innovative integration of AI with petroleum engineering principles not only refines prediction accuracy but also establishes a foundation for autonomous operational adjustments, paving the way for self-optimizing production systems (Kuang et al., 2021). The synergy between AI-driven models and real-time operational data offers an unprecedented opportunity to address the challenges of water loading, ensuring sustained production and optimized resource management in the upstream geoenergy industry (Li et al., 2024) (Bello et al., 2016).

Literature Review

Traditional Methods for Predicting Water Loading

Historically, the prediction of water loading in gas wells has relied heavily on empirical models derived from simplified physical principles and extensive field observations. The pioneering work in liquid loading prediction was conducted by Turner et al. in 1969 .Their research focused on calculating the critical gas velocity required to prevent liquid accumulation. They studied a dataset of 106 vertical wells, primarily with 2.5-inch pipe diameters and wellhead pressures exceeding 500 psia.

Their investigations led to the development of two conceptual models: a continuous film model and an entrained drop movement model. Through experimentation, the entrained drop model, which conceptualizes a force balance on a single liquid droplet, was found to provide the best fit to their field data . This model, while foundational, possesses significant limitations, including its inability to account for varying pipe diameters, inclination angles, or the complex multiphase flow dynamics observed in modern offshore wells (Shekhar et al., 2017) (Yusuf et al., 2010).

Turner's Model and its Limitations

Despite its widespread use, Turner's model often overestimates or underestimates critical flow rates due to its simplistic assumptions and failure to incorporate the complex interactions of multiphase flow, varying fluid properties, and wellbore geometries (Dousi et al., 2005). Specifically, the model struggles with accurately predicting critical gas rates in deviated or horizontal wells, where gravitational effects and flow patterns deviate significantly from the vertical flow assumptions upon which the model was built. This limitation is particularly pronounced in unconventional reservoirs and offshore environments, where complex well trajectories and highly variable operating conditions are common (Shekhar & Kelkar, 2016).

Furthermore, the model does not adequately account for liquid holdup in horizontal sections, which can significantly impair production even before liquid loading becomes evident in the production tubing (Denney, 2012). The model's inability to dynamically adapt to changes in fluid properties, such as gas gravity or water-cut variations, further restricts its applicability in wells with evolving production characteristics (Michel & Civan, 2008). Moreover, Turner's model, primarily developed for gas wells, encounters substantial inaccuracies when applied to wells with significant oil production or multiphase flow streams, necessitating the development of more comprehensive models that incorporate the complex interplay of oil, gas, and water (He & Wang, 2014).

AI Applications in Gas Well Optimization

The advent of artificial intelligence and machine learning has revolutionized various industries, including oil and gas, by offering advanced capabilities for data analysis, pattern recognition, and predictive modeling, which are particularly valuable for optimizing gas well performance and mitigating issues like water loading (Srinivasan et al., 2021). These computational advancements provide an alternative to traditional empirical and mechanistic models, which often fall short in complex, high-dimensional datasets characteristic of petroleum engineering (Abdrakhmanov et al., 2021). AI and machine learning applications in the oil and gas industry include precision drilling, automation, and real-time monitoring of wells, contributing to substantial time and cost savings (Tariq et al., 2021). Specifically, AI and machine learning techniques can be applied to enhance production optimization, minimize downtime, improve safety protocols, and advance exploration and drilling methodologies (Li et al., 2024).

Such sophisticated analytical tools enable the identification of subtle patterns and correlations within vast operational datasets, leading to more accurate predictions of well behavior and optimized intervention strategies (Li et al., 2024) (Azmi et al., 2024).

Data-Driven Approaches for Predicting Critical Gas Rates

The application of data-driven approaches, particularly artificial intelligence and machine learning, has gained significant traction in predicting critical gas rates, offering a robust alternative to conventional methods by leveraging the increasing availability of high-resolution operational data (Tariq et al., 2021). These methods excel at uncovering intricate, non-linear relationships within complex datasets that traditional physics-based models often overlook, leading to more accurate and adaptable predictions (Hosseini & Akilan, 2023). This shift toward data-centric methodologies, including soft computing and machine learning, has recently garnered significant interest and acceptance within the oil and gas industry for various applications, including fault prediction and overall system optimization (Nybø, 2010).

By analyzing historical production data, real-time sensor readings, and well parameters, AI models can learn to predict the onset of liquid loading with greater precision, enabling proactive intervention strategies (Gupta et al., 2018) (Tariq et al., 2021). This capability allows operators to transition from reactive maintenance, where equipment fails before being addressed, to proactive maintenance, where potential failures are predicted and mitigated before they occur (Chen et al., 2020). This paradigm shift enables the optimization of production strategies, reduction of operational costs, and enhancement of overall well productivity. Furthermore, these advanced AI models can integrate diverse data sources, such as geological surveys, drilling logs, and fluid analyses, to build a holistic understanding of reservoir dynamics and wellbore conditions, leading to superior predictive accuracy and more effective mitigation strategies.

The capacity of these models to process and interpret "big data"—high volumes of diverse data types at rapid speeds—is pivotal for transforming raw operational data into actionable insights, thereby accelerating decision-making and improving overall efficiency (Feblowitz, 2013) (Hoffmann et al., 2020).

Methodology

This section outlines the systematic methodology developed to create an AI- enhanced, optimized empirical model for predicting critical gas rates, thereby addressing the deficiencies of established techniques such as Turner's model.

The historical evolution of these predictive models illustrates a progression from fundamental assumptions to more sophisticated attempts to incorporate complex physical phenomena, including flow regimes, droplet deformation, liquid holdup, and bottomhole conditions. This ongoing effort to refine the underlying physical principles underscores the inherent complexity of multiphase flow within dynamic wellbores.

Nevertheless, even the most advanced conventional models, exemplified by those from Guo and Luan & He, still present limitations, such as the omission of specific physical terms or reduced predictive accuracy under particular conditions of liquid loading. This persistent difficulty in perfectly encapsulating all physical nuances through explicit mathematical formulations suggests that a universally applicable analytical solution remains elusive. Consequently, this situation provides a strong justification for investigating data-driven methodologies, such as deep learning, which possess the potential to implicitly account for these intricate physical interactions and phenomena not explicitly modeled.

A notable trend observed across numerous conventional models is the explicit specification of "applicability ranges," often correlated with wellhead pressure. For instance, Turner's revised model and Coleman's model are discussed in the context of wellhead pressures, as are the models proposed by Nosseir, Wang & Liu, Zhou & Yuan, and Luan & He. Although Guo's model is presented as having "no limitation" in its applicability, it still exhibits minor inaccuracies within specific flow regimes. This fragmentation of applicability implies that no single conventional model demonstrates universal robustness across the full spectrum of gas well operating conditions. Consequently, operators frequently must select distinct models based on their well's unique characteristics, which introduces complexity and the potential for errors if an inappropriate model is chosen. This pervasive limitation underscores a critical necessity for a predictive tool exhibiting broader generalizability.

Furthermore, the development and validation of these models are substantially contingent upon empirical data. Turner's initial model was derived from data encompassing 106 wells; however, issues such as "16 are questionable" and "lack of some well data" were identified. Subsequent models have often involved re-analyzing Turner's dataset or utilizing their own, sometimes limited, new data. The explicit statement that calculations were not performed for all wells due to "lack of some well data in Turner's original set and being unwilling to make assumptions for them, as it could result in errors" clearly illustrates the direct impact of data quality and completeness on the development and validation processes of these models.

This establishes a feedback loop where the refinement of a model is inherently constrained by the availability of appropriate data. The accuracy and generalizability of any predictive model, particularly those that are data-driven, are fundamentally limited by the quality, completeness, and diversity of the training dataset. This highlights that robust data acquisition, cleaning, and management strategies are as critical as the model architecture itself.

Data Acquisition and Preprocessing

To address these challenges, this study focused on compiling a comprehensive, high-resolution dataset from multiple, representative well, encompassing critical operational parameters such as bottomhole pressure, wellhead pressure, gas properties, and production rates. Variance analysis and permutation entropy were subsequently applied to this dataset to precisely detect early-stage flow regime transitions indicative of liquid loading onset. Polynomial feature transformations were then employed to capture complex nonlinear and cross- variable interactions among key operational parameters, enhancing the model's ability to discern subtle dependencies crucial for accurate prediction.

Rigorous preprocessing steps were implemented to mitigate noise and handle outliers within the data, ensuring the integrity and reliability of the dataset for subsequent model training (Sapronova & Marcher, 2025). This meticulous data preparation phase is crucial for developing a robust predictive model that can generalize across diverse operational conditions, especially considering the challenges of limited and heterogeneous data in the petroleum industry (Alfarraj & AlRegib, 2019) (Weinbrandt et al., 1998).

Feature Engineering and Selection

Building upon this foundation, feature engineering involved the meticulous creation of new variables from the raw measurements, leveraging domain expertise to encapsulate complex physical relationships pertinent to multiphase flow dynamics. This process included deriving dimensionless groups and interaction terms that reflect the interplay between fluid properties, well geometry, and operational settings, significantly enhancing the predictive power of the model. These engineered features, such as the comprehensive set of input parameters selected based on their relative importance to the output, were then subjected to rigorous selection criteria to identify the most salient predictors, thereby optimizing computational efficiency without compromising model accuracy (Mustafa et al., 2023).

The generation of features, a critical step, involved using domain knowledge to extract meaningful attributes from the raw data (Ivlev, 2021). This approach significantly improved the model's capacity to interpret the subtle, often non-linear, relationships governing liquid loading phenomena, particularly when dealing with varying well settings and controls (Badawi & Gildin, 2024). For instance, polynomial feature transformations, including second-order and interaction terms between flowing bottomhole pressure and gas density (ρg), were strategically applied to capture intricate nonlinear behaviors characteristic of multiphase flow in offshore environments (Heaton, 2016).

Model Development

The application of these sophisticated feature engineering techniques directly facilitated the development of a highly accurate predictive model, capable of discerning critical gas rates with minimal deviation from observed values. This enabled the model to accurately capture complex nonlinear behaviors in multiphase environments, confirming its robustness across varying well conditions. The flexible structure of this AI-based model allows seamless integration into digital twin platforms and real-time surveillance systems, enabling dynamic monitoring and proactive optimization of deliquification strategies (Yuan & Bello, 2014). This integration transforms raw data into actionable intelligence through automated "smart flows," assisting engineers in daily well surveillance and improving decision-making processes for enhanced asset performance (Al-Jasmi et al., 2013).

This framework leverages a synergistic combination of data-driven insights and physics-based understanding, thereby providing a comprehensive and robust toolkit for managing complex well behaviors and optimizing production in dynamic offshore environments (Xiao et al., 2018). Furthermore, the system's capacity to process massive quantities of data in real-time allows for the conversion of raw sensor outputs into meaningful operational insights, facilitating timely and informed interventions (Al-Jasmi et al., 2013). This predictive capability extends to supporting long-term field development plans by optimizing well targeting, identifying infill opportunities, and enhancing water-flood management strategies (Hwang et al., 2019). Moreover, the seamless integration of this model into existing production workflows and digital twin platforms offers significant advancements in operational efficiency and predictive maintenance capabilities (Su et al., 2016).

Variance Analysis and Permutation Entropy

In the context of this study, variance analysis was employed to quantify the dispersion or spread of data points around the mean for critical parameters, thereby identifying variables exhibiting significant fluctuations indicative of changing flow regimes. Concurrently, permutation entropy provided a robust, model-free measure of the complexity and predictability of time series data, allowing for the early detection of subtle, non-linear patterns associated with the incipient stages of liquid loading (Xu et al., 2021). These analytical techniques collectively enabled a more precise and timely identification of transitions, which is crucial for proactive management of deliquification strategies and preventing catastrophic well failures (Jeong et al., 2020).

Such sophisticated diagnostic tools enhance the precision of predictive models by furnishing high-fidelity inputs that accurately reflect the complex hydrodynamics within the wellbore, thereby supporting proactive interventions (Adeyinka et al., 2020). This comprehensive approach ensures that the developed AI model is not only statistically robust but also physically informed, offering unparalleled accuracy in predicting critical gas rates and optimizing production in complex offshore environments.

Polynomial Feature Transformations

The strategic application of polynomial feature transformations, particularly involving second-order and interaction terms, was instrumental in capturing the intricate non-linear relationships between various operational parameters, such as flowing bottomhole pressure and gas density (Wang et al., 2019). This technique allowed the model to accurately represent the complex interplay of these variables in multiphase flow environments, leading to a more precise prediction of critical gas rates. This methodological enhancement significantly improved the model's ability to discern subtle changes in fluid dynamics that precede liquid loading, offering a substantial advantage over traditional linear models. This sophisticated approach thereby enabled the development of a highly accurate predictive model, capable of discerning critical gas rates with minimal deviation from observed values (Blinstrub et al., 2014).

The flexible structure of this AI-based model allows seamless integration into digital twin platforms and real-time surveillance systems, enabling dynamic monitoring and proactive optimization of deliquification strategies.

Supervised Learning Framework

This framework, specifically utilizing polynomial regression, achieved a remarkable 92% accuracy in predicting critical gas rates, showcasing a substantial 28% improvement over conventional Turner-based methodologies (Lai, 2025). This significant advancement underscores the efficacy of machine learning in addressing complex engineering problems, particularly in scenarios where traditional empirical models fall short due to their inherent limitations in capturing nonlinearities (Liu et al., 2013) (Castillo et al., 2021). The supervised learning paradigm, by enabling the model to learn directly from high-resolution, real-field datasets, provided a robust mechanism for capturing the subtle interdependencies and dynamic behaviors within the well system (Wei et al., 2017). This data-driven approach, leveraging advanced regression techniques, offers a novel and more accurate means of forecasting production and optimizing operational parameters, particularly in challenging environments like those encountered in offshore oil and gas extraction (Hosseini & Akilan, 2023) (Hong et al., 2020).

Model Validation and Performance Metrics

The model's robustness was confirmed through rigorous cross-validation techniques and extensive comparison against actual field performance data, demonstrating its consistent predictive accuracy across diverse operating conditions. Minimal deviation between predicted and actual critical gas rates was observed, further validating the model's reliability for real-world applications and supporting its use for optimizing reservoir delivery performance (Schulze-Riegert et al., 2020). The ability to accurately forecast field behavior and production from existing wells, based on modifications of operational constraints, positions this model as a valuable complementary tool to conventional numerical simulators for complex assets (Masoudi et al., 2020). Furthermore, the interpretability of this advanced model, despite its complexity, allows domain experts to understand and diagnose the underlying physical phenomena, bridging the gap between data-driven predictions and established engineering principles (Liu & Pyrcz, 2023). This synergistic integration of data- driven insights with foundational engineering knowledge significantly enhances decision-making processes for gas well management.

Results and Discussion

This section presents the empirical findings from the AI-enhanced prediction model, critically analyzing its performance against traditional methods and discussing the implications for gas well management. It details the significant improvements in prediction accuracy and adaptability demonstrated by the novel Turner-style equation, emphasizing its potential to revolutionize proactive deliquification strategies. The discussion will also highlight the model's capacity for seamless integration into existing digital twin platforms and real-time surveillance systems, showcasing its utility for continuous optimization of production systems (Wong et al., 2018) (Dilib et al., 2013).

The model's superior performance stems from its ability to dynamically adapt to varying well conditions, a critical advantage over static conventional models. Specifically, the polynomial regression approach, incorporating second-order and interaction terms, enables the model to accurately capture complex nonlinear behaviors in multiphase offshore environments, overcoming limitations associated with linear assumptions. This adaptability is crucial for maintaining optimal production and preventing liquid loading in dynamic well environments.

The enhanced model demonstrates superior accuracy in predicting critical gas rates, exhibiting a marked improvement over conventional Turner models, which often fail to account for the dynamic intricacies of multiphase flow (Arinze et al., 2024). This improvement is largely attributable to the model's sophisticated integration of machine learning algorithms, which allows for a more nuanced interpretation of high-resolution well datasets (Srikonda et al., 2020). This innovative approach leverages advanced analytics to identify subtle patterns and correlations within the data that are often missed by traditional methods, thereby refining the predictive capability for critical flow conditions (Hosseini & Akilan, 2023).

Comparison with Traditional Turner-Based Methods

The AI-enhanced model significantly outperforms traditional Turner-based methods, which are inherently limited by their empirical derivation from simplified assumptions and their inability to adapt to the complex, non-linear dynamics of real-world multiphase flow environments (Hong et al., 2020). This enhanced adaptability stems from its capacity to continuously learn and refine its predictions based on new data, a feature absent in static conventional models.

Analysis of Key Parameters

The model's superior performance is attributed to its ability to identify and quantify the influence of various operational and physical parameters, such as bottomhole pressure and gas density, on critical gas rates through polynomial feature transformations. These transformations effectively capture the intricate, nonlinear relationships and cross-variable interactions that dictate the onset of liquid loading, providing a more comprehensive understanding than previously possible with linear models. The meticulous analysis of these parameters provides unprecedented insights into the complex hydrodynamics within gas wells, facilitating more informed decision-making for preventative measures and optimized production strategies. This advanced analytical capability enables proactive adjustments to operational parameters, leading to sustained production efficiency and reduced intervention costs.

Model Accuracy and Robustness

The model consistently achieved 92% accuracy in predicting critical gas rates, representing a significant 28% improvement over conventional Turner-based methods, and demonstrating robustness across varying well conditions and multiphase flow regimes. This high level of accuracy and broad applicability underscores the model's potential as a reliable tool for real-time deliquification management and strategic production optimization in the petroleum industry. Furthermore, its capacity to accurately forecast critical flow conditions under a variety of operational scenarios positions it as an indispensable asset for proactive well management. The model's ability to retain information from previous patterns and behaviors, akin to a memory, allows it to provide informed decisions, even with incomplete knowledge of the underlying physical system (Alakeely & Horne, 2020).

Real-Time Adaptability

The flexible architecture of the AI-based model facilitates seamless integration into digital twin platforms and real-time surveillance systems, enabling continuous monitoring and dynamic optimization of production strategies (Cullick & Sukkestad, 2010). This integration provides operators with immediate insights into well performance, allowing for rapid adjustments to prevent liquid loading and maintain optimal gas production. This adaptability is critical for proactive intervention, minimizing downtime and maximizing resource recovery in complex offshore environments.

Integration with Digital Twin Platforms

The integration into digital twin platforms provides a virtual replica of the physical well, enabling operators to simulate various scenarios and predict the impact of different operational changes before implementing them in the field. This capability enhances decision-making by offering a predictive environment for testing hypotheses, thereby optimizing production and mitigating risks in a cost-effective manner (Aladsani et al., 2019). This synergistic relationship between the AI model and digital twins revolutionizes asset integrity management, allowing personnel to shift focus from labor-intensive assessments to applying outcomes for enhanced system understanding (Renzi et al., 2017).

Conclusion

This study successfully developed an AI-enhanced, optimized empirical model for predicting critical gas rates, showcasing significant advancements over traditional Turner-style equations through the integration of real-field data and advanced machine learning techniques. The model dynamically adapts to varying well conditions, outperforming traditional methods and enabling proactive, real-time management of deliquification strategies in complex environments. High-resolution well datasets, including bottomhole pressure, wellhead pressure, gas properties, and production rates, were extracted from a well X and rigorously preprocessed to ensure data quality and relevance for model training. Variance analysis and permutation entropy were applied to detect early-stage flow regime transitions associated with liquid loading onset, allowing for a more accurate representation of dynamic well behavior.

Implications for Gas Production

The developed AI model, by accurately predicting critical gas rates, has profound implications for optimizing gas production and mitigating liquid loading challenges in unconventional reservoirs. It offers a robust data-driven alternative to conventional methods by leveraging machine learning to capture complex, non-linear relationships that influence multiphase flow dynamics (Chen et al., 2025) (Sircar et al., 2021). This allows for more precise and timely interventions, ultimately leading to enhanced operational efficiency and substantial cost savings in the upstream oil and gas sector (Koroteev & Tekić, 2020) (Gowekar, 2024).

Limitations of the Study

While the model demonstrates significant improvements, its current iteration is primarily based on data from a single well, which may limit its immediate generalizability across diverse geological formations and operational conditions without further calibration. Future research will focus on validating the model across a broader spectrum of wells and integrating additional well-specific parameters to enhance its universal applicability. This includes incorporating data from wells with varying production histories and geological characteristics to ensure the model's predictive accuracy across different operational environments (Sircar et al., 2021). Furthermore, the reliance on historical data, while extensive, means the model's performance could be impacted by unforeseen operational shifts or the emergence of novel flow behaviors not represented in the training dataset, necessitating continuous model retraining and adaptation.

Future Research Directions

Future work will also explore the integration of unsupervised learning techniques to identify subtle patterns in operational data that might indicate nascent loading conditions even before traditional thresholds are met, thus pushing the boundaries of predictive maintenance. Further research will also explore the development of hybrid models that combine physics-based simulations with machine learning algorithms to achieve even greater accuracy and interpretability, particularly in scenarios involving complex multiphase flow (Aghito & Bjørkevoll, 2020). The extension of this research to include real-time optimization algorithms will enable autonomous well management, where critical parameters are adjusted automatically to prevent liquid loading and optimize production. Moreover, investigating the model's performance with multi-phase flow, beyond just single-phase or two-phase systems, would be beneficial for addressing real-world complexities in oil fields (Xu et al., 2021) (Chen et al., 2009).

Additionally, incorporating a wider array of sensor data, such as distributed acoustic sensing or fiber optic temperature sensing, could provide richer datasets for training more sophisticated AI models, further enhancing predictive capabilities for complex wellbore phenomena. Beyond the technical enhancements, future studies should also delve into the economic implications of deploying such AI-driven models, quantifying the return on investment from reduced intervention costs and increased hydrocarbon recovery. The development of such AI-driven systems represents a significant step towards the autonomous and optimized operation of gas wells, paving the way for more sustainable and efficient energy production practices (Sircar et al., 2021).

References

Abdrakhmanov, I. R., Kanin, E., Бороhиh, C. A., Burnaev, E., & Osiptsov, A. A. (2021). Development of Deep Transformer-Based Models for Long-Term Prediction of Transient Production of Oil Wells. SPE Russian Petroleum Technology Conference. https://doi.org/10.2118/206537-ms

Adeyinka, A., Tsakporhore, A., Oduwole, A., Jumbo, S., Odusote, F., Aliyev, S., Adegoke, O., Lapham, J., & Eme, T. (2020). Re-positioning Mature Fields for Sustainability: Deriving Additional Value by Applying the Internet of Things via Remote Well Monitoring. SPE Nigeria Annual International Conference and Exhibition. https://doi.org/10.2118/203671-ms

Aghito, M., & Bjørkevoll, K. S. (2020). Hybrid Approach for Drilling Automation. Day 3 Tue, November 30, 2021. https://doi.org/10.2118/200734-ms

Ajayi, A., Fassihi, R. R., Wrobel, A., Allouache, A., Johnson, M. D., Bryant, D., & Akajagbor, C. (2019, September 18). Application of Multiphase-Multilayer Pressure Transient Analysis for Well Monitoring and Reservoir Management - A Case Study of Deep-Water Offshore GoM Reservoirs. SPE Annual Technical Conference and Exhibition. https://doi.org/10.2118/196161-ms

Aladsani, A. Y., Alghafli, A. H., Kaabi, S. H. A., McNeilly, K., Akhtar, M. M., Tripathi, D., Alkuwaiti, H., Soni, S., & Isambertt, J. (2019, November 8). Actual Well Performance Identification and Production Efficiency Enhancement and Sustainability in a Brown Field. Day 3 Wed, November 17, 2021. https://doi.org/10.2118/197383-ms

Alakeely, A., & Horne, R. N. (2020). Simulating the Behavior of Reservoirs with Convolutional and Recurrent Neural Networks. *SPE Reservoir Evaluation & Engineering*, **23**(3), 992. https://doi.org/10.2118/201193-pa

AlBar, A., Asfoor, H., Goz, A., & Ansari, N. (2019). Combining the Power of IoT and Big Data to Unleash the Potential of Digital Oil Field. https://doi.org/10.2523/iptc-19045-ms

Alfarraj, M., & AlRegib, G. (2019). Petrophysical Property Estimation from Seismic Data Using Recurrent Neural Networks. arXiv (Cornell University). https://doi.org/10.48550/arxiv.1901.08623

Al-Jasmi, A. ., Nasr, H. N., Goel, H. K., Moricca, G.. , Carvajal, G. A., Dhar, J. ., Querales, M. ., Villamizar, M.., Cullick, A. S., García, J. A. R., Velasquez, G. ., Yong, Z., Bermudez, F. ., & Khain, J.. (2013). ESP "Smart Flow" Integrates Quality and Control Data for Diagnostics and Optimization in Real Time (Part of KwIDF Project). Day 2 Tue, October 04, 2022. https://doi.org/10.2118/167394-ms

Arinze, C. A., Izionworu, V. O., Isong, D. E., Daudu, C. D., & Adefemi, A. (2024). Integrating artificial intelligence into engineering processes for improved efficiency and safety in oil and gas operations. *Open Access Research Journal of Engineering and Technology*, **6**(1), 39. https://doi.org/10.53022/oarjet.2024.6.1.0012

Azmi, P. A. R., Yusoff, M., & Sallehud-din, M. T. M. (2024). A Review of Predictive Analytics Models in the Oil and Gas Industries [Review of A Review of Predictive Analytics Models in the Oil and Gas Industries]. *Sensors*, **24**(12), 4013. Multidisciplinary Digital Publishing Institute. https://doi.org/10.3390/s24124013

Badawi, D., & Gildin, E. (2024). Neural operator-based proxy for reservoir simulations considering varying well settings, locations, and permeability fields. *Computers & Geosciences*, 105826. https://doi.org/10.1016/j.cageo.2024.105826

Bello, O., Teodoriu, C., Yaqoob, T., Oppelt, J., Holzmann, J., & Obiwanne, A. (2016). Application of Artificial Intelligence Techniques in Drilling System Design and Operations: A State of the Art Review and Future Research Pathways. SPE Nigeria Annual International Conference and Exhibition. https://doi.org/10.2118/184320-ms

Blinstrub, J., Li, Y. G., Newby, M., Zhou, Q., Stigant, G., Pilidis, P., & Hönen, H. (2014). Application of Gas Path Analysis to Compressor Diagnosis of an Industrial Gas Turbine Using Field Data. https://doi.org/10.1115/gt2014-25330

Bolen, M., Crkvenjakov, V., & Converset, J. (2018). The Role of Big Data in Operational Excellence and Real Time Fleet Performance Management—The Key to Deepwater Thriving in a Low-Cost Oil Environment. IADC/SPE Drilling Conference and Exhibition. https://doi.org/10.2118/189603-ms

Cadei, L., Montini, M., Landi, F., Porcelli, F., Michetti, V., Origgi, M., Tonegutti, M., & Duranton, S. (2018, November 10). Big Data Advanced Anlytics to Forecast Operational Upsets in Upstream Production System. Day 3 Tue, November 30, 2021. https://doi.org/10.2118/193190-ms

Castillo, I. G., Loboda, I., & Pérez-Ruiz, J. L. (2021). Data-Driven Models for Gas Turbine Online Diagnosis. *Machines*, **9**(12), 372. https://doi.org/10.3390/machines9120372

Castiñeira, D., Zhai, X., Darabi, H., Valle, M. del, Maqui, A., Shahvali, M., & Yunuskhojayev, A. (2018). Augmented AI Solutions for Heavy Oil Reservoirs: Innovative Workflows That Build from Smart Analytics, Machine Learning And Expert-Based Systems. https://doi.org/10.2118/193650-ms

Chen, F., Sun, L., Jian, S., Huo, X., Pan, X., Feng, C., & Zhang, Z. (2025). A Review of AI Applications in Unconventional Oil and Gas Exploration and Development [Review of A Review of AI Applications in Unconventional Oil and Gas Exploration and Development]. *Energies*, **18**(2), 391. Multidisciplinary Digital Publishing Institute. https://doi.org/10.3390/en18020391

Chen, Y., Bernath, A., Rai, H., & Muron, P. (2009). Development and Application of Upscaling Techniques for Modeling Near-Well Flow in Heterogeneous Reservoirs. SPE Annual Technical Conference and Exhibition. https://doi.org/10.2118/124955-ms

Chen, Z., Selere, Olugbenro. O., & Seng, N. L. C. (2020). Equipment Failure Analysis for Oil and Gas Industry with an Ensemble Predictive Model. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2012.15030

Chernikov, A. D., Eremin, N. A., Stolyarov, V. E., Sboev, A., Semenova-Chashchina, O. K., & Fitsner, L. K. (2020). Application of artificial intelligence methods for identifying and predicting complications in the construction of oil and gas wells: problems and solutions. *Georesursy*, **22**(3), 87. https://doi.org/10.18599/grs.2020.3.87-96

Cui, E. (2015). Detailed Reservoir Characterization for Key Strategies of Unconventional Reservoir Development. SPE Reservoir Characterisation and Simulation Conference and Exhibition. https://doi.org/10.2118/175628-ms

Cullick, A. S., & Sukkestad, T. (2010, March 23). Smart Operations with Intelligent Well Systems. SPE Intelligent Energy Conference and Exhibition. https://doi.org/10.2118/126246-ms

D'Almeida, A. L., Bergiante, N. C. R., Ferreira, G. de S., Leta, F. R., Lima, C. B. de C., & Lima, G. B. A. (2022). Digital transformation: a review on artificial intelligence techniques in drilling and production applications [Review of Digital transformation: a review on artificial intelligence techniques in drilling and production applications]. *The International Journal of Advanced Manufacturing Technology*, **119**, 5553. Springer Science+Business Media. https://doi.org/10.1007/s00170-021-08631-w

Deif, A. M., & Vivek, T. (2022). Understanding AI Application Dynamics in Oil and Gas Supply Chain Management and Development: A Location Perspective. *HighTech and Innovation Journal*, **3**, 1. https://doi.org/10.28991/hij-sp2022-03-01

Denney, D. (2012). Investigating Liquid Loading in Tight Horizontal Gas Wells. *Journal of Petroleum Technology*, **64**(11), 138. https://doi.org/10.2118/1112-0138-jpt

Dilib, F. A., Jackson, M. D., & Khairullin, A. Y. (2013, September 30). Field Production Optimization Using Closed-Loop Direct Feedback Control of Intelligent Wells: Application to the Brugge Model. SPE Annual Technical Conference and Exhibition. https://doi.org/10.2118/166384-ms

Dousi, N., Veeken, C., & Currie, P. K. (2005, September 6). Modelling the gas well liquid loading process. *All Days*. https://doi.org/10.2118/95282-ms

Feblowitz, J. (2013). Analytics in Oil and Gas: The Big Deal About Big Data. *All Days*. https://doi.org/10.2118/163717-ms

Gandhi, Y., Zheng, K., Jha, B., Nomura, K., Nakano, A., Vashishta, P., & Kalia, R. K. (2024). Generative AI- driven forecasting of oil production. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2409.16482

Golenkin, M. Y., Bulygin, I. A., Shapovalov, A. P., Zavyalov, A. A., Zhirkina, A. A., Pico, Y. L., & Knyazev, I. V. (2020). Digital Production Optimization Boost for North Caspian Field With 3-Phase Real-Time Downhole Flow Metering System From Enhanced Intelligent Completion. SPE Russian Petroleum Technology Conference. https://doi.org/10.2118/201993-ms

Gowekar, G. S. (2024a). Artificial intelligence for predictive maintenance in oil and gas operations. *World Journal of Advanced Research and Reviews*, **23**(3), 1228. https://doi.org/10.30574/wjarr.2024.23.3.2721

Gowekar, G. S. (2024b). How oil and gas industry are transforming with AI and ML. *World Journal of Advanced Research and Reviews*, **23**(3), 1234. https://doi.org/10.30574/wjarr.2024.23.3.2722

Gupta, S., Sharma, A., & Abubakar, A. (2018, September 24). Artificial Intelligence–Driven Asset Optimizer. SPE Annual Technical Conference and Exhibition. https://doi.org/10.2118/191551-ms

Hajizadeh, Y. (2019). Machine learning in oil and gas; a SWOT analysis approach. *Journal of Petroleum Science and Engineering*, **176**, 661. https://doi.org/10.1016/j.petrol.2019.01.113

He, F. Y., & Wang, F. (2014). Research on Energy Saving for Multi-Back after Blending Water from One Station System's Parameters Optimization. *Advanced Materials Research*, **1023**, 187. https://doi.org/10.4028/www.scientific.net/amr.1023.187

Heaton, J. (2016). An empirical analysis of feature engineering for predictive modeling. *SoutheastCon*, 1. https://doi.org/10.1109/secon.2016.7506650

Hingerl, F., Arnst, B., Cosby, D., Kreutzman, L., & Tyree, R. (2020). The Future of Plunger Lift Control Using Artificial Intelligence. Day 3 Tue, November 30, 2021. https://doi.org/10.2118/201132-ms

Hoffmann, M., Wildermuth, S., Gitzel, R., Boyaci, A., Gebhardt, J., Kaul, H., Amihai, I., Forg, B., Suriyah, M., Leibfried, T., Stich, V., Hicking, J., Bremer, M., Kaminski, L., Beverungen, D., Heiden, P. zur, & Tornede, T. (2020). Integration of Novel Sensors and Machine Learning for Predictive Maintenance in Medium Voltage Switchgear to Enable the Energy and Mobility Revolutions [Review of Integration of Novel Sensors and Machine Learning for Predictive Maintenance in Medium Voltage Switchgear to Enable the Energy and Mobility Revolutions]. *Sensors*, **20**(7), 2099. Multidisciplinary Digital Publishing Institute. https://doi.org/10.3390/s20072099

Hong, L., Yu, H., Cao, N., Tian, H., & Cheng, S. (2020). Applications of Artificial Intelligence in Oil and Gas Development. *Archives of Computational Methods in Engineering*, **28**(3), 937. https://doi.org/10.1007/s11831-020-09402-8

Hossain, M., & Ismail, M. D. bin M. (2013). Potential Application of Downhole Gas Compressor to Improve Productivity for Gas Reservoir. *All Days*. https://doi.org/10.2523/iptc-16982-ms

Hosseini, S., & Akilan, T. (2023). Advanced Deep Regression Models for Forecasting Time Series Oil Production. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2308.16105

Hwang, Y. K., Zirczy, H., & Bakku, S. K. (2019). Quantitative seismic reservoir modeling — Model-based probabilistic inversion for optimal field development. *The Leading Edge*, **38**(10), 786. https://doi.org/10.1190/tle38100786.1

Ivlev, D. (2021). Prediction of geophysical properties of rocks on rare well data and attributes of seismic waves by machine learning methods on the example of the Achimov formation. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2106.13274

Jeong, C., Yu, Y., Mansour, D., Vesslinov, V., & Meehan, R. (2020). A Physics Model Embedded Hybrid Deep Neural Network for Drillstring Washout Detection. IADC/SPE International Drilling Conference and Exhibition. https://doi.org/10.2118/199629-ms

Koroteev, D., & Tekić, Ž. (2020). Artificial intelligence in oil and gas upstream: Trends, challenges, and scenarios for the future. *Energy and AI*, **3**, 100041. https://doi.org/10.1016/j.egyai.2020.100041

Kuang, L., Liu, H., Ren, Y., Luo, K., Shi, M., Su, J., & Li, X. (2021). Application and development trend of artificial intelligence in petroleum exploration and development. *Petroleum Exploration and Development*, **48**(1), 1. https://doi.org/10.1016/s1876-3804(21)60001-0

Lai, J.-S. (2025). Research on prediction algorithm of effluent quality and development of integrated control system for waste-water treatment. *Scientific Reports*, **15**(1). https://doi.org/10.1038/s41598-025-03612-5

Lehr, D. (2020). Technology Focus: Completions (April 2020). *Journal of Petroleum Technology*, **72**(4), 51. https://doi.org/10.2118/0420-0051-jpt

Li, J. X., Zhang, T., Zhu, Y., & Chen, Z. (2024a). Artificial General Intelligence (AGI) for the oil and gas industry: a review [Review of Artificial General Intelligence (AGI) for the oil and gas industry: a review]. arXiv (Cornell University). Cornell University. https://doi.org/10.48550/arxiv.2406.00594

Li, J. X., Zhang, T., Zhu, Y., & Chen, Z. (2024b). Artificial general intelligence for the upstream geoenergy industry: a review [Review of Artificial general intelligence for the upstream geoenergy industry: a review]. *Gas Science and Engineering*, **131**, 205469. Elsevier BV. https://doi.org/10.1016/j.jgsce.2024.205469

Linden, R. J., Marck, J. W., & Boer, J. P. de. (2015). Results of Real-Time Production Optimization of a maturing North Sea Gas Asset with Production Constraints. *IFAC-PapersOnLine*, **48**(6), 45. https://doi.org/10.1016/j.ifacol.2015.08.008

Liu, W., & Pyrcz, M. J. (2023). Physics-informed graph neural network for spatial-temporal production forecasting. *Geoenergy Science and Engineering*, **223**, 211486. https://doi.org/10.1016/j.geoen.2023.211486

Liu, Y., & Jiang, X. (2022). Towards Predictive Maintenance of a Heavy-Duty Gas Turbine A New Hybrid Intelligent Methodology for Performance Simulation. *Annual Conference of the PHM Society*, **14**(1). https://doi.org/10.36001/phmconf.2022.v14i1.3148

Liu, Y., Yao, K.-T., Raghavenda, C. S., Wu, A., Guo, D., Zheng, J., Olabinjo, L., Balogun, O., & Ershaghi, I. (2013). Global Model for Failure Prediction for Rod Pump Artificial Lift Systems. https://doi.org/10.2118/165374-ms

Marmier, R., Decroux, B., Paran, M.-A., Raurich, S., Akhundov, H., & Esor, E. (2017). Unlocking the Development of Absheron Gas Condensate Field through an Integrated Asset Work and Modeling (Russian). SPE Annual Caspian Technical Conference and Exhibition. https://doi.org/10.2118/188995-ru

Masoudi, R., Mohaghegh, S. D., Yingling, D., Ansari, A. H., Amat, H., Mohamad, N., Sabzabadi, A., & Mandel, D. (2020). Subsurface Analytics Case Study; Reservoir Simulation and Modeling of Highly Complex Offshore Field in Malaysia, Using Artificial Intelligent and Machine Learning. SPE Annual Technical Conference and Exhibition. https://doi.org/10.2118/201693-ms

Michel, G., & Civan, F. (2008). Modeling Nonisothermal Rapid Multiphase Flow in Wells Under Nonequilibrium Conditions. *SPE Production & Operations*, **23**(2), 125. https://doi.org/10.2118/102231-pa

Mustafa, A., Tariq, Z., Mahmoud, M., & Abdulraheem, A. (2023). Machine learning accelerated approach to infer nuclear magnetic resonance porosity for a middle eastern carbonate reservoir. *Scientific Reports*, **13**(1). https://doi.org/10.1038/s41598-023-30708-7

Nybø, R. (2010). Fault detection and other time series opportunities in the petroleum industry. *Neurocomputing*, **73**, 1987. https://doi.org/10.1016/j.neucom.2009.10.020

Pandey, Y. N., Rastogi, A., Kainkaryam, S., Bhattacharya, S., & Saputelli, L. (2020). Opportunities, Challenges, and Future Trends. In *Apress eBooks* (p. 259). https://doi.org/10.1007/978-1-4842-6094-4_8

Pennel, M., Hsiung, J., & Putcha, V. B. (2018). Detecting Failures and Optimizing Performance in Artificial Lift Using Machine Learning Models. *SPE Western Regional Meeting*. https://doi.org/10.2118/190090-ms

Renzi, D., Maniar, D., McNeill, S., & Vecchio, C. D. (2017). Developing a Digital Twin for Floating Production Systems Integrity Management. https://doi.org/10.4043/28012-ms

Sapronova, A., & Marcher, T. (2025). Improving Data Quality with Advanced Pre-Processing of MWD Data. *Geotechnics*, **5**(2), 28. https://doi.org/10.3390/geotechnics5020028

Saputelli, L., Nikolaou, M., & Economides, M. J. (2005). Self-Learning Reservoir Management. *SPE Reservoir Evaluation & Engineering*, **8**(6), 534. https://doi.org/10.2118/84064-pa

Sarma, P., Fowler, G., Henning, M., Kyriacou, S., Thakur, G., Sloss, D., & Shipley, W. (2017). Cyclic Steam Injection Modeling and Optimization for Candidate Selection, Steam Volume Optimization, and SOR Minimization, Powered by Unique, Fast, Modeling and Data Assimilation Algorithms. *SPE Western Regional Meeting*. https://doi.org/10.2118/185747-ms

Schulze-Riegert, R., Lang, P., Pongtepupathum, W., Drew, C., Round, A. J., Shaw, K., Adeyemi, T., Topdemir, S., Pattie, S., Nasiri, H., & Hegre, T. M. (2020). Ensemble-Based Well Location Optimization Under Subsurface Uncertainty Guided By Deep-Learning Approach To 3D Geological Feature Classification. Day 3 Wed, November 17, 2021. https://doi.org/10.2118/202660-ms

Shekhar, S., & Kelkar, M. (2016, September 21). Prediction of Onset of Liquid Loading in Vertical, Inclined and Near Horizontal Wells. Day 3 Wed, October 05, 2022. https://doi.org/10.2118/181244-ms

Shekhar, S., Kelkar, M., Hearn, W. J., & Hain, L. L. (2017). Improved Prediction of Liquid Loading In Gas Wells. *SPE Production & Operations*, **32**(4), 539. https://doi.org/10.2118/186088-pa

Shen, F., Ren, S., Zhang, X. Y., Luo, H., & Feng, C. (2021). A Digital Twin-Based Approach for Optimization and Prediction of Oil and Gas Production. *Mathematical Problems in Engineering*, 2021, 1. https://doi.org/10.1155/2021/3062841

Sircar, A., Yadav, K., Rayavarapu, K., Bist, N., & Oza, H. (2021). Application of machine learning and artificial intelligence in oil and gas industry. *Petroleum Research*, **6**(4), 379. https://doi.org/10.1016/j.ptlrs.2021.05.009

Srikonda, R., Rastogi, A., & Oestensen, H. (2020). Increasing Facility Uptime Using Machine Learning and Physics-Based Hybrid Analytics in a Dynamic Digital Twin. https://doi.org/10.4043/30723-ms

Srinivasan, S., O'Malley, D., Mudunuru, M. K., Sweeney, M., Hyman, J. D., Karra, S., Frash, L., Carey, J. W., Gross, M. R., Guthrie, G., Carr, T. R., Li, L., & Viswanathan, H. (2021). A machine learning framework for rapid forecasting and history matching in unconventional reservoirs. *Scientific Reports*, **11**(1). https://doi.org/10.1038/s41598-021-01023-w

Su, S., Patacchini, L., Mohmed, F., Farouk, M., Ouzzane, D., Draoui, E., Torrens, R., & Amoudruz, P. (2016). Coupling Production and Injection Systems with Multiple Reservoir Models: A Novel Method of Optimizing Development Strategies in a Mature Giant Oilfield. https://doi.org/10.2118/183153-ms

Syed, F. I., Alshamsi, M., Dahaghi, A. K., & Neghabhan, S. (2020). Artificial lift system optimization using machine learning applications. *Petroleum*, **8**(2), 219. https://doi.org/10.1016/j.petlm.2020.08.003

Tariq, Z., Aljawad, M. S., Hasan, A., Murtaza, M., Mohammed, E. A., El-Husseiny, A., Alarifi, S. A., Mahmoud, M., & Abdulraheem, A. (2021). A systematic review of data science and machine learning applications to the oil and gas industry [Review of A systematic review of data science and machine learning applications to the oil and gas industry]. *Journal of Petroleum Exploration and Production Technology*, **11**(12), 4339. Springer Nature. https://doi.org/10.1007/s13202-021-01302-2

Temizel, C., Nabizadeh, M., Kadkhodaei, N., Ranjith, R., Suhag, A., Balaji, K., & Dhannoon, D. (2017, May 9). Data-Driven Optimization of Injection/Production in Waterflood Operations. *All Days*. https://doi.org/10.2118/187468-ms

Wang, H., Mu, L., Fugeng, S., Liu, K., & Qian, Y. (2019). Management and instant query of distributed oil and gas production dynamic data. *Petroleum Exploration and Development*, **46**(5), 1014. https://doi.org/10.1016/s1876-3804(19)60258-2

Wang, J., Luo, X., Zhang, X., & Du, S. (2025). Artificial General Intelligence (AGI) Applications and Prospect in Oil and Gas Reservoir Development. *Processes*, **13**(5), 1413. https://doi.org/10.3390/pr13051413

Waqar, A., Othman, I., Shafiq, N., & Mansoor, M. S. (2023). Applications of AI in oil and gas projects towards sustainable development: a systematic literature review. *Artificial Intelligence Review*, **56**(11), 12771. https://doi.org/10.1007/s10462-023-10467-7

Wei, C., Wang, Y., Ding, Y., Li, Y., Shi, J., Liu, S., Tian, C., Li, B., Xiong, L., & Zhang, Q. (2017). Uncertainty Assessment in Production Forecasting and Optimization for a Giant Multi-Layered Sandstone Reservoir Using Optimized Artificial Neural Network Technology. SPE Reservoir Characterisation and Simulation Conference and Exhibition. https://doi.org/10.2118/186069-ms

Weinbrandt, R. M., Trentham, R. C., & Robinson, W. (1998). Incorporating Seismic Attribute Porosity into a Flow Model of the Grayburg Reservoir. *SPE/DOE Improved Oil Recovery Symposium*. https://doi.org/10.2118/39666-ms

Wong, J. C. Y., Ghous, A., Morgan, J., & Hourani, M. Y. (2018, October 17). Realising Additional Value from Late Life High Water Loaded Gas Wells through Transient Flow Modelling. SPE Asia Pacific Oil and Gas Conference and Exhibition. https://doi.org/10.2118/192009-ms

Xiao, F., Long, T., Asokan, B. V., El-Bakry, A., Dani, N. R., Holub, C., & Poole, S. (2018). A Robust Surveillance and Optimization System for Unstable Flows in Offshore Gas Lifted Wells. https://doi.org/10.2118/193144-ms

Xu, R., Zhang, D., & Wang, N. (2021). Uncertainty quantification and inverse modeling for subsurface flow in 3D heterogeneous formations using a theory-guided convolutional encoder-decoder network. arXiv (Cornell University). https://doi.org/10.48550/arxiv.2111.08691

Yuan, K., & Bello, O. (2014, April 1). Use of Computational Fluid Dynamics Model for Evaluating Performance of High Pressure High Temperature Wells. *All Days*. https://doi.org/10.2118/167896-ms

Yusuf, R., Veeken, K., & Hu, B. (2010, January 20). Investigation of Gas Well Liquid Loading With a Transient Multiphase Flow Model. SPE Oil and Gas India Conference and Exhibition. https://doi.org/10.2118/128470-ms

Zhao, R., Shi, J., Zhang, X., Li, J., Peng, Y., Xiong, C., Liu, M., Deng, F., Song, S., & Miao, G. (2019). Research and Application of the Big Data Analysis Platform of Oil and Gas Production. https://doi.org/10.2523/iptc-19458-ms